

## CS306 PHASE 3

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### *Description*

In phase 3, I have applied CRUD operations to the “Teams” and “Players” tables that I have used previously in phase 2. I have inserted new values through python to these tables and then showed both that query was succesful and checked through each step by looking at the relational database in mySQL. Then continued with updating already existing rows their certain attributes. I updated “team\_name” from teams and “transfer\_value” from players tables. Then I deleted certain rows and showed it through both python output and mySQL relational table.

## *First look of tables*

These are the tables teams and players before doing any operations

The screenshot shows the Trendyol Superleague Explorer interface. The top bar includes navigation icons and tabs for Administration, Schemas, Query 1, SQL File 3\*, and SQL File 13\*. The left sidebar shows a Schemas tree with SYS and TFF folders. The main area displays a SQL query with four lines:

```
1 select * from Teams;
2 select * from Players;
3 select * from manager;
4 select * from plays_at;
```

Below the query, the Result Grid shows a table with 10 rows and 2 columns: Team\_ID and Team\_Name. The data is as follows:

| Team_ID | Team_Name      |
|---------|----------------|
| 1       | Fenerbahçe     |
| 2       | Galatasaray    |
| 3       | Beykötaş       |
| 4       | Trabzonspor    |
| 5       | Antalyaspor    |
| 6       | Kayserispor    |
| 7       | Adana Demirsor |
| 8       | Kasimpasa      |
| 9       | Samsunspor     |
| 10      | Konyaspor      |

The bottom of the interface shows the Action Output section with a single entry:

| Time | Action                                      | Response           | Duration / Fetch Time  |
|------|---------------------------------------------|--------------------|------------------------|
| 10   | 20:43:11 select * from Teams LIMIT 0, 50000 | 10 row(s) returned | 0.00039 sec / 0.000... |

Teams

The screenshot shows the Trendyol Superleague Explorer interface. The SQL query editor contains the following code:

```

1 * select * from Teams;
2 * select * from Players;
3 * select * from managers;
4 * select * from plays_at;

```

The results are displayed in a table with the following data:

| Player_ID | Player_Name        | Transfer_value |
|-----------|--------------------|----------------|
| 100       | Edin Dzeko         | 5.5            |
| 101       | Mauro Icardi       | 8              |
| 102       | Al Musrati         | 3.5            |
| 103       | Enis Bardhi        | 4.5            |
| 104       | Adam Buksa         | 1              |
| 105       | Majid Hosseini     | 2              |
| 106       | Mario Balotelli    | 1.8            |
| 107       | Haris Hajradinovic | 2.2            |
| 108       | Carlo Hoes         | 2.75           |
| 109       | Sokol Cikalleshi   | 1              |

The bottom status bar shows: Action Output 11 20:44:13 select \* from Players LIMIT 0, 50000. Response: 10 row(s) returned. Duration / Fetch Time: 0.00043 sec / 0.000...

## Players

### *1.Insert Operation*

I inserted for teams and players and below pictures I showed that query was succesful and gave relational database after insertion. I provided the functions I used and succesfull queries and also relational database after insertion.

## *Function for insertion*

```
# Insert Operation
def insert_teams(connection, data):
    query = "INSERT INTO Teams VALUES (%s,%s)"
    print(query)
    execute_query(connection, query, data)

def insert_players(connection, data):
    query = "INSERT INTO Players VALUES (%s,%s,%s)"
    print(query)
    execute_query(connection, query, data)
```

```
18
19 # Insert Operation
20 def insert_teams(connection, data):
21     query = "INSERT INTO Teams VALUES (%s,%s)"
22     print(query)
23     execute_query(connection, query, data)
24
25
26 values_to_insert = [
27     (11, "Ankaragücü"),
28     (12, "İstanbulspor"),
29     (13, "Bajaksheir")
30 ]
31
32 fields = {
33     "Team_ID": "INT PRIMARY KEY",
34     "Team_Name": "VARCHAR(250)"
35 }
36
37
38 connection = create_connection()
39
40 for team in values_to_insert:
41     insert_teams(connection,team)
```

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```
INSERT INTO Teams VALUES (%s,%s)
Query executed successfully
INSERT INTO Teams VALUES (%s,%s)
Query executed successfully
INSERT INTO Teams VALUES (%s,%s)
Query executed successfully
Eralp-MacBook-Air:cs386 project eralpkipiziloglu$
```

bash  
Python

*Teams succesfull query*

The screenshot shows the Trendyol Superleague Explorer interface. On the left, a sidebar lists database objects: Administration, Schemas, and Tables. The 'Tables' section is expanded, showing a list of tables including Management, Manager, Players, Plays\_at, Teams, Columns, Indexes, Foreign Keys, Triggers, Views, Stored Procedures, and Functions. The 'Teams' table is selected. The main area displays a SQL query: `1 select * from Teams; 2 select * from Players; 3 select * from manager; 4 select * from plays_at;`. Below the query, the 'Result Grid' shows the data for the 'Teams' table. The grid has two columns: 'Team\_ID' and 'Team\_Name'. The data is as follows:

| Team_ID | Team_Name       |
|---------|-----------------|
| 1       | Fenerbahçe      |
| 2       | Galatasaray     |
| 3       | Beşiktaş        |
| 4       | Trabzonspor     |
| 5       | Antalyaspor     |
| 6       | Kayserispor     |
| 7       | Adana Demirspor |
| 8       | Kasımpaşa       |
| 9       | Samsunspor      |
| 10      | Konyaspor       |
| 11      | Ankaragücü      |
| 12      | Istanbulspor    |
| 13      | Bağakşehir      |

At the bottom, the 'Action Output' section shows the query execution details: `select * from Teams LIMIT 0, 50000`. The response indicates that 13 rows were returned, and the duration was 0.00037 sec / 0.0000...

## Teams Table

```

47 values_to_insert_players = [
48     (110, "Tolga Cigerci",1.5),
49     (111, "Racine Co'ly",1.1),
50     (112, "Dimitris Pelkas",2.5)
51 ]
52
53 field_players = {
54     "Player_ID": "INT PRIMARY KEY",
55     "Player_Name": "VARCHAR(50)",
56     "Transfer_value" : "FLOAT"
57 }
58
59
60 connection = create_connection()
61
62 #for team in values_to_insert:
63 #    insert_teams(connection,team)
64
65
66 for player in values_to_insert_players:
67     insert_players(connection,player)
68

```

The terminal output shows the successful execution of the queries:

```

Connection established with the database
INSERT INTO Players VALUES (%s,%s,%s)
Query executed successfully
INSERT INTO Players VALUES (%s,%s,%s)
Query executed successfully
INSERT INTO Players VALUES (%s,%s,%s)
Query executed successfully

```

## Players succesful query



## *2.Read Operation*

Here this part I tested if I was able to read the tables through python application on the output and created a function for it.

### *Function for reading*

```
#READTABLES
def read_tables(connection,table_name):
    query = f"SELECT * FROM `{table_name}`"
    cursor = connection.cursor(dictionary=True)
    try:
        cursor.execute(query)
        records = cursor.fetchall()
        for record in records:
            print(record)
    except Error as e:
        print(f"Error: {e}")
    finally:
        cursor.close()
```



## Reading teams

```
115 read_tables(connection,"Teams")
```

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● Eralp-MacBook-Air:cs306 project eralpkiziloglu\$ /usr/local/bin/python3 "/Users/eralpkiziloglu/Downloads/cs306 project/tff.py"

Connection established with the database

```
{'Team_ID': 1, 'Team_Name': 'Fenerbahce'}
{'Team_ID': 2, 'Team_Name': 'Galatasaray'}
{'Team_ID': 3, 'Team_Name': 'Besiktas'}
{'Team_ID': 4, 'Team_Name': 'Trabzonspor'}
{'Team_ID': 5, 'Team_Name': 'Antalyaspor'}
{'Team_ID': 6, 'Team_Name': 'Kayserispor'}
{'Team_ID': 7, 'Team_Name': 'Adana Demirspor'}
{'Team_ID': 8, 'Team_Name': 'Kasimpasa'}
{'Team_ID': 9, 'Team_Name': 'Samsunspor'}
{'Team_ID': 10, 'Team_Name': 'Konyaspor'}
{'Team_ID': 11, 'Team_Name': 'Ankaragucu'}
{'Team_ID': 12, 'Team_Name': 'Istanbulspor'}
{'Team_ID': 13, 'Team_Name': 'Basaksehir'}
```

○ Eralp-MacBook-Air:cs306 project eralpkiziloglu\$

Ln 41, Col 44 Spaces: 4 UTF-8 LF Python 3.12.1 64-bit

## Reading Players

```
118 read_tables(connection,"Players")
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

● Eralp-MacBook-Air:cs306 project eralpkiziloglu\$ /usr/local/bin/python3 "/Users/eralpkiziloglu/Downloads/cs306 project/tff.py"

Connection established with the database

```
{'Player_ID': 100, 'Player_Name': 'Edin Dzeko', 'Transfer_value': 5.5}
{'Player_ID': 101, 'Player_Name': 'Mauro icardi', 'Transfer_value': 8.0}
{'Player_ID': 102, 'Player_Name': 'Al Musrati', 'Transfer_value': 3.5}
{'Player_ID': 103, 'Player_Name': 'Enis Bardhi', 'Transfer_value': 4.5}
{'Player_ID': 104, 'Player_Name': 'Adam Buksa', 'Transfer_value': 1.0}
{'Player_ID': 105, 'Player_Name': 'Majid Hosseini', 'Transfer_value': 2.0}
{'Player_ID': 106, 'Player_Name': 'Mario Balotelli', 'Transfer_value': 1.8}
{'Player_ID': 107, 'Player_Name': 'Haris Hajradinovic', 'Transfer_value': 2.2}
{'Player_ID': 108, 'Player_Name': 'Carlo Holse', 'Transfer_value': 2.75}
{'Player_ID': 109, 'Player_Name': 'Sokol Cikalleshi', 'Transfer_value': 1.0}
{'Player_ID': 110, 'Player_Name': 'Tolga Cigerci', 'Transfer_value': 1.5}
{'Player_ID': 111, 'Player_Name': 'Racine Coly', 'Transfer_value': 1.1}
{'Player_ID': 112, 'Player_Name': 'Dimitris Pelkas', 'Transfer_value': 2.5}
```

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### *3.Update Operation*

Here I update certain attribute values of rows where it is team\_name for teams and transfer\_value for players also provided sql tables version for it.

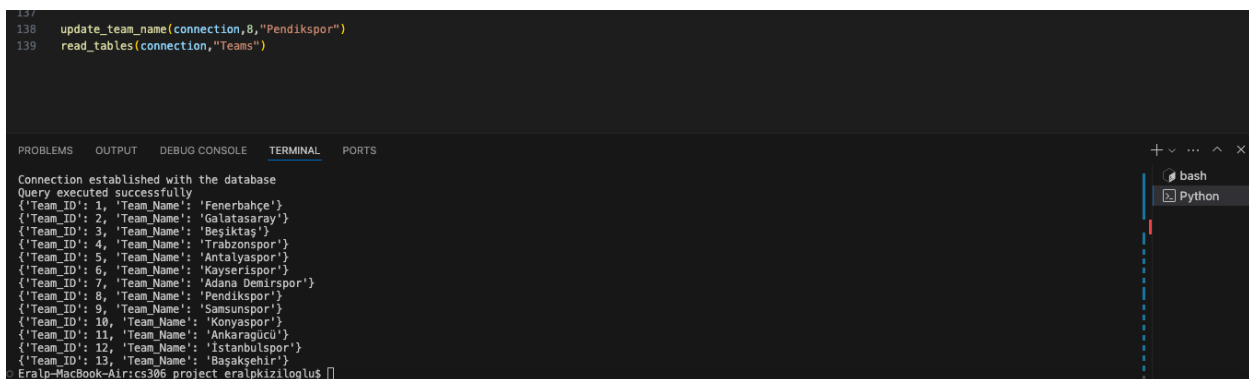
#### *Function for updating*

```
def update_team_name(connection, team_id, new_team_name):
    query = "UPDATE Teams SET Team_Name = %s WHERE team_ID = %s"
    data = (new_team_name,team_id)
    execute_query(connection, query, data)

def update_player_value(connection, player_ID, new_transfer_value):
    query = "UPDATE Players SET transfer_value = %s WHERE player_ID = %s"
    data = (new_transfer_value,player_ID)
    execute_query(connection, query, data)
```

#### *Update succesful for teams*

```
137
138 update_team_name(connection,8,"Pendikspor")
139 read_tables(connection,"Teams")
```



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Connection established with the database  
Query executed successfully  
{'Team\_ID': 1, 'Team\_Name': 'Fenerbahce'}  
{'Team\_ID': 2, 'Team\_Name': 'Galatasaray'}  
{'Team\_ID': 3, 'Team\_Name': 'Besiktas'}  
{'Team\_ID': 4, 'Team\_Name': 'Trabzonspor'}  
{'Team\_ID': 5, 'Team\_Name': 'Antalyaspor'}  
{'Team\_ID': 6, 'Team\_Name': 'Kayserispor'}  
{'Team\_ID': 7, 'Team\_Name': 'Adana Demirspor'}  
{'Team\_ID': 8, 'Team\_Name': 'Pendikspor'}  
{'Team\_ID': 9, 'Team\_Name': 'Samsunspor'}  
{'Team\_ID': 10, 'Team\_Name': 'Konyaspor'}  
{'Team\_ID': 11, 'Team\_Name': 'Ankaragucu'}  
{'Team\_ID': 12, 'Team\_Name': 'Istanbulspor'}  
{'Team\_ID': 13, 'Team\_Name': 'Basaksehir'}

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## Sql Table for teams after update

The screenshot shows the Trendyol Superleague Explorer interface. The left sidebar displays the database schema with the 'Teams' table selected. The main area shows the SQL query: `1. select * from Teams;`. The 'Result Grid' displays the following data:

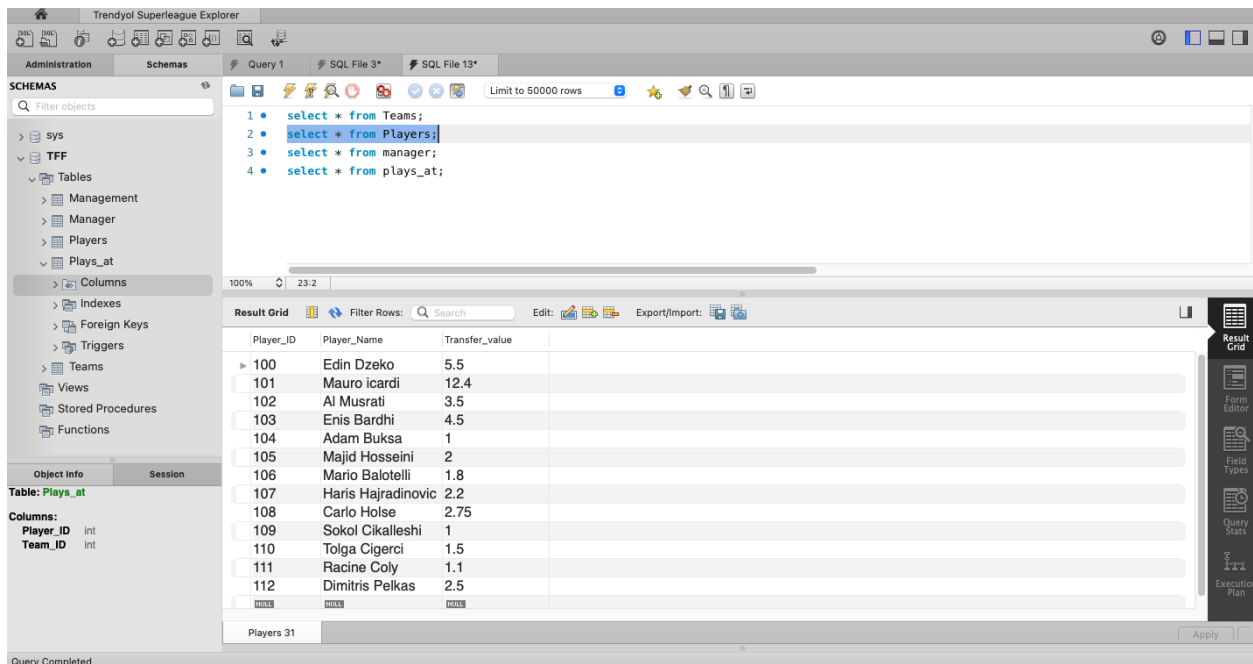
| Team_ID | Team_Name       |
|---------|-----------------|
| 1       | Fenerbahçe      |
| 2       | Galatasaray     |
| 3       | Beşiktaş        |
| 4       | Trabzonspor     |
| 5       | Antalyaspor     |
| 6       | Kayserispor     |
| 7       | Adana Demirspor |
| 8       | Pendikspor      |
| 9       | Samsunspor      |
| 10      | Konyaspor       |
| 11      | Ankaragücü      |
| 12      | İstanbulspor    |
| 13      | Başakşehir      |

## Update Successful for players

The screenshot shows a terminal window with the following output:

```
139 update_player_value(connection,101,12.4)
140 read_tables(connection,"players")

Connection established with the database
Query executed successfully
{'Player_ID': 100, 'Player_Name': 'Edin Dzeko', 'Transfer_value': 5.5}
{'Player_ID': 101, 'Player_Name': 'Mauro Icardi', 'Transfer_value': 12.4}
{'Player_ID': 102, 'Player_Name': 'Al Musrati', 'Transfer_value': 3.5}
{'Player_ID': 103, 'Player_Name': 'Enis Bardhi', 'Transfer_value': 4.5}
{'Player_ID': 104, 'Player_Name': 'Adam Buksa', 'Transfer_value': 1.0}
{'Player_ID': 105, 'Player_Name': 'Majid Hosseini', 'Transfer_value': 2.0}
{'Player_ID': 106, 'Player_Name': 'Mario Balotelli', 'Transfer_value': 1.8}
{'Player_ID': 107, 'Player_Name': 'Maris Hajradinovic', 'Transfer_value': 2.2}
{'Player_ID': 108, 'Player_Name': 'Carlo Haise', 'Transfer_value': 2.75}
{'Player_ID': 109, 'Player_Name': 'Sokol Cikalleshi', 'Transfer_value': 1.0}
{'Player_ID': 110, 'Player_Name': 'Tolga Cigerci', 'Transfer_value': 1.5}
{'Player_ID': 111, 'Player_Name': 'Racine Coly', 'Transfer_value': 1.1}
{'Player_ID': 112, 'Player_Name': 'Dimitris Pelkas', 'Transfer_value': 2.5}
Eralp-MacBook-Air:cs306 project eralpkiziloglu$
```



## 4.Delete Operation

Here I deleted certain rows from my tables that existed and again shared function also query being succesful and at the end I shared sql versions of the final versions overall.

### Function for deleting

```
# Delete tables rows
def delete_team(connection, Team_ID):
    query = "DELETE FROM teams WHERE Team_ID = %s"
    data = (Team_ID,)
    execute_query(connection, query, data)

def delete_player(connection, player_ID):
    query = "DELETE FROM players WHERE player_ID = %s"
    data = (player_ID,)
    execute_query(connection, query, data)
```

## Successful delete query teams

```
189 delete_team(connection,8)
190 delete_team(connection,11)
191 read_tables(connection,"Teams")
192
193
194
195
196
```

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{'Team\_ID': 10, 'Team\_Name': 'Konyaspor'}  
{'Team\_ID': 11, 'Team\_Name': 'Ankaragücü'}  
{'Team\_ID': 12, 'Team\_Name': 'İstanbulspor'}  
{'Team\_ID': 13, 'Team\_Name': 'Başakşehir'}

● Eralp-MacBook-Air:cs306 project eralpkiziloglu\$ /usr/local/bin/python3 "/Users/eralpkiziloglu/Downloads/cs306 project/tff.py"  
Connection established with the database  
Query executed successfully  
Query executed successfully  
{'Team\_ID': 1, 'Team\_Name': 'Fenerbahçe'}  
{'Team\_ID': 2, 'Team\_Name': 'Galatasaray'}  
{'Team\_ID': 3, 'Team\_Name': 'Beşiktaş'}  
{'Team\_ID': 4, 'Team\_Name': 'Trabzonspor'}  
{'Team\_ID': 5, 'Team\_Name': 'Antalyaspor'}  
{'Team\_ID': 6, 'Team\_Name': 'Kayserispor'}  
{'Team\_ID': 7, 'Team\_Name': 'Adana Demirspor'}  
{'Team\_ID': 9, 'Team\_Name': 'Samsunspor'}  
{'Team\_ID': 10, 'Team\_Name': 'Konyaspor'}  
{'Team\_ID': 12, 'Team\_Name': 'İstanbulspor'}  
{'Team\_ID': 13, 'Team\_Name': 'Başakşehir'}  
○ Eralp-MacBook-Air:cs306 project eralpkiziloglu\$

## Successful delete query players

```
194 #delete_player(connection,109)
195 read_tables(connection,"Players")
196
197
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

{'Team\_ID': 13, 'Team\_Name': 'Başakşehir'}

● Eralp-MacBook-Air:cs306 project eralpkiziloglu\$ /usr/local/bin/python3 "/Users/eralpkiziloglu/Downloads/cs306 project/tff.py"  
Connection established with the database  
Query executed successfully  
Query executed successfully  
Error: 1146 (42502): Table 'tff.player' doesn't exist  
● Eralp-MacBook-Air:cs306 project eralpkiziloglu\$ /usr/local/bin/python3 "/Users/eralpkiziloglu/Downloads/cs306 project/tff.py"  
Connection established with the database  
{'Player\_ID': 100, 'Player\_Name': 'Edin Dzeko', 'Transfer\_value': 5.5}  
{'Player\_ID': 101, 'Player\_Name': 'Mauro icardi', 'Transfer\_value': 12.4}  
{'Player\_ID': 102, 'Player\_Name': 'Al Musrati', 'Transfer\_value': 3.5}  
{'Player\_ID': 103, 'Player\_Name': 'Enis Bardhi', 'Transfer\_value': 4.5}  
{'Player\_ID': 105, 'Player\_Name': 'Majid Hosseini', 'Transfer\_value': 2.0}  
{'Player\_ID': 106, 'Player\_Name': 'Mario Balotelli', 'Transfer\_value': 1.8}  
{'Player\_ID': 107, 'Player\_Name': 'Maris Hajradinovic', 'Transfer\_value': 2.2}  
{'Player\_ID': 108, 'Player\_Name': 'Carlo Holse', 'Transfer\_value': 2.75}  
{'Player\_ID': 110, 'Player\_Name': 'Tolga Cigerci', 'Transfer\_value': 1.5}  
{'Player\_ID': 111, 'Player\_Name': 'Racine Coly', 'Transfer\_value': 1.1}  
{'Player\_ID': 112, 'Player\_Name': 'Dimitris Pelkas', 'Transfer\_value': 2.5}  
○ Eralp-MacBook-Air:cs306 project eralpkiziloglu\$

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## Final version of tables

Trendyol Superleague Explorer

Administration Schemas Query 1 SQL File 3\* SQL File 13\*

Limit to 50000 rows

```
1 select * from Teams;
2 select * from Players;
3 select * from manager;
4 select * from plays_at;
```

Result Grid

| Team_ID | Team_Name       |
|---------|-----------------|
| 1       | Fenerbahçe      |
| 2       | Galatasaray     |
| 3       | Beşiktaş        |
| 4       | Trabzonspor     |
| 5       | Antalyaspor     |
| 6       | Kayserispor     |
| 7       | Adana Demirspor |
| 9       | Samsunspor      |
| 10      | Konyaspor       |
| 12      | İstanbulspor    |
| 13      | Başakşehir      |

Teams 2

Action Output

| Time     | Action  | Response          | Duration / Fetch Time |
|----------|---------|-------------------|-----------------------|
| 23:55:44 | use TFF | 0 row(s) affected | 0.00028 sec           |

Trendyol Superleague Explorer

Administration Schemas Query 1 SQL File 3\* SQL File 13\*

Limit to 50000 rows

```
1 select * from Teams;
2 select * from Players;
3 select * from manager;
4 select * from plays_at;
```

Result Grid

| Player_ID | Player_Name        | Transfer_value |
|-----------|--------------------|----------------|
| 100       | Edin Dzeko         | 5.5            |
| 101       | Mauro icardi       | 12.4           |
| 102       | Al Musrati         | 3.5            |
| 103       | Enis Bardhi        | 4.5            |
| 105       | Majid Hosseini     | 2              |
| 106       | Mario Balotelli    | 1.8            |
| 107       | Haris Hajradinovic | 2.2            |
| 108       | Carlo Holse        | 2.75           |
| 110       | Tolga Cigerci      | 1.5            |
| 111       | Racine Coly        | 1.1            |
| 112       | Dimitris Pelkas    | 2.5            |

Players 3

Action Output

| Time     | Action                               | Response           | Duration / Fetch Time   |
|----------|--------------------------------------|--------------------|-------------------------|
| 23:58:25 | select * from Players LIMIT 0, 50000 | 11 row(s) returned | 0.00061 sec / 0.0000... |